

mum as it completes the desired number of rotations.

At this time, if S5 is at position 1 (repeat) and input is high at PB3 pin, the motor will continue to rotate in clockwise and anticlockwise directions at selected speeds. But if S5 is at position 2 (single) and input is low, the motor will stop after one rotation. After two seconds, you will be prompted to enter the desired number of rotations for the next operation, and the cycle will continue.

## Software

ATmega16 is the main building block of this project because it performs the following tasks:

1. Takes user input from pushbutton switches to set speed, rotations, run motor and so on.
2. Rotates the motor at the desired speed using PWM technique.
3. Displays motor speed, rotations and various messages on the LCD.
4. Counts actual motor rotations.
5. Automatically reverses the motor when it completes the desired number of rotations.
6. Rotates the motor clockwise or anticlockwise in repeat mode.

To implement all the above functionalities, the software program is loaded into the internal ROM (flash) of the microcontroller. The program (DCMotor\_speed.c) is written in C language and is compiled using AVR Simulator 2 available with AVR Studio IDE software. The hex code generated is used to burn into ATmega16 using a suitable programmer. (ProgISP programmer was used at EFM Lab for programming the hex code.)

## Construction and testing

An actual-size, single-size PCB layout for the auto-reversible DC motor control is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit, enclose it in a suitable box.

Fix CON1 at the back side of the cabinet and S1 through S5 at the top side of the cabinet. Fix S6 on the front side of the cabinet for resetting the circuit. Attach a light thin metal strip to the shaft of motor in such a way that when shaft rotates the metal strip should pass through the slot (cut) of the MOC7811.

After assembling the circuit on the PCB, cross-check for any wrong connections. Burn the program (DCMotor\_speed.hex) into the microcontroller using a suitable programmer.

It is recommended to use a 40-pin IC base for ATmega16. Use a 5V DC power supply for the circuit and a 12V DC power supply for the DC motor. **EFY**



Ashutosh M. Bhatt is M.Tech in embedded systems. Currently, he is a lecturer of electronics and computer engineering at Government Polytechnic, Jamnagar, Gujarat



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